

Arjuna JEE 2.0 2027

Maths

Basic Mathematics

DPP: 5

Total Duration - 40 Min

Q1 Values of 'x' satisfying both inequalities:

$$x - 3 \leq 4 \text{ \& } \frac{x-5}{x+1} > 0$$

- (A) $x \in (-\infty, 7]$
 (B) $x \in (-\infty, -1)$
 (C) $x \in (-\infty, -1) \cup (5, 7]$
 (D) $x \in (5, 7]$

Q2 Solution of inequality $2x - 3 > 0$ and

$$4 - 3x < 0$$

- (A) $x > \frac{4}{3}$
 (B) $x < \frac{4}{3}$
 (C) $x > \frac{3}{2}$
 (D) $x < \frac{3}{2}$

Q3 All real set of values of x satisfying

$$x^2 - 3x + 2 > 0 \text{ and } x^2 - 2x - 4 \leq 0 \text{ are given by } [a, b] \cup (c, d] \text{ then value of } \frac{a+d}{c-b} \text{ is equal to } \underline{\hspace{2cm}}.$$

Q4 Number of integer values of x satisfying

$$5x - 1 < (x + 1)^2 < 7x - 3 \text{ are}$$

Q5 Values of 'x' satisfying: $-1 < \frac{2x}{x-1} \leq 4$

- (A) $x \in (-\infty, \frac{1}{3}) \cup (1, \infty)$
 (B) $x \in (-\infty, \frac{1}{3}) \cup (2, \infty)$
 (C) $x \in (-\infty, \frac{1}{3}) \cup [2, \infty)$
 (D) $x \in (-\infty, \frac{1}{3}) \cup [2, \infty) \cup \{1\}$

Q6 Solution of $(x - 1)^2(x + 1)^3(x - 4) \geq 0$ is

- (A) $x \in (-\infty, -1] \cup \{2\} \cup [4, \infty)$
 (B) $x \in (-\infty, 1] \cup \{3\} \cup [4, \infty)$
 (C) $x \in (-\infty, -1] \cup \{1\} \cup [4, \infty)$
 (D) $x \in (-\infty, -1] \cup \{1\} \cup [3, \infty)$

Q7 Solution of $\frac{x^4(x+1)^2(x-2)}{(x-3)^3(x+4)} > 0$ is

- (A) $x \in (-4, 2) \cup (4, \infty)$
 (B) $x \in (-4, -1) \cup (-1, 2) \cup (3, \infty)$
 (C) $x \in (-4, 0) \cup (0, 2) \cup (3, \infty)$

$$(D) x \in (-4, -1) \cup (-1, 0) \cup (0, 2) \cup (3, \infty)$$

Q8 The complete solution set of the inequality

$$\frac{x^4 - 3x^3 + 2x^2}{x^2 - x - 30} \geq 0$$

- (A) $(-\infty, -5) \cup (1, 2) \cup (6, \infty) \cup \{0\}$
 (B) $(-\infty, -5) \cup [1, 2] \cup (6, \infty) \cup \{0\}$
 (C) $(-\infty, -5) \cup [1, 2] \cup [6, \infty)$
 (D) None of these

Q9 Solution of $(x^2 + 1) > 0$ is

- (A) $x \in (-\infty, -1) \cup (1, \infty)$
 (B) $x \in (-1, 1)$
 (C) $x \in \mathbb{R}$
 (D) No solution

Q10 Complete set of values of 'x' :

$$\frac{(x^2 - 11x + 24)^{101}}{(x^2 - 10x + 24)^{100}} \leq 0$$

- (A) $x \in [3, 8]$
 (B) $x \in [3, 4) \cup (6, 8]$
 (C) $x \in [3, 8] - \{4, 6\}$
 (D) $x \in (-\infty, 3] \cup [8, \infty)$

Q11 The solution set for

$$x(x+2)^2(x-1)^5(2x-3)(x-3)^4 \geq 0 \text{ is given by } x \in [a, b] \cup [c, \infty) \cup \{-2\} \text{ then value of } a + b + c \text{ is equal to } \underline{\hspace{2cm}}.$$

Q12 The number of integral solution of $\frac{x+1}{x^2+2} > \frac{1}{4}$ is

Q13 Complete set of values of 'x' satisfying:

$$\frac{(x-1)(x^2-x+12)}{(3+x^2)} \leq 0$$

- (A) $x \geq 1$
 (B) $x \in (-\infty, -3] \cup (-\sqrt{3}, 1] \cup (\sqrt{3}, 4]$
 (C) $x \in (-\infty, -3] \cup [1, 4]$
 (D) $x \leq 1$

Q14 Complete set of values of 'x' satisfying:

$$\frac{(x-1)^{15}x^{28}}{(x+1)^{2021}(x+2)^{2020}} \geq 0$$



- (A) $x \in (-\infty, -1) \cup [1, \infty)$
(B) $x \in (-1, 1] - \{0\}$
(C) $x \in (-\infty, -2] \cup (0, \infty)$
(D) $x \in (-\infty, -2) \cup (-2, -1) \cup [1, \infty) \cup \{0\}$

Q15 Solve $\frac{x^2+2x-3}{x^2+1} < 0$

Q16 Solve $\frac{(x-1)(x+2)^2}{-1-x} < 0$

Q17 Solve $x^4 - 2x^2 - 63 \leq 0$

Q18 Solve $\frac{x+1}{(x-1)^2} < 1$

Q19 Solve $\frac{x^2-7x+12}{2x^2+4x+5} > 0$

Q20 solve $\frac{x^2+6x-7}{x^2+1} \leq 2$

Q21 Solve $\frac{1}{x+2} < \frac{3}{x-3}$

Q22 Solve $\frac{(2-x^2)(x-3)^3}{(x+1)(x^2-3x-4)} \geq 0$



Answer Key

Q1 (C)
Q2 (C)
Q3 2
Q4 1
Q5 (C)
Q6 (C)
Q7 (D)
Q8 (B)
Q9 (C)
Q10 (C)
Q11 2.5

Q12 5
Q13 (D)
Q14 (D)
Q15 $x \in (-3, 1)$
Q16 $x \in (-\infty, -2) \cup (-2, -1) \cup (1, \infty)$
Q17 $x \in [-3, 3]$
Q18 $x \in (-\infty, 0) \cup (3, \infty)$
Q19 $(-\infty, 3) \cup (4, +\infty)$
Q20 $x \in \mathbb{R}$
Q21 $x \in \left(-\frac{9}{2}, -2\right) \cup (3, \infty)$
Q22 $x \in [-\sqrt{2}, -1) \cup (-1, \sqrt{2}] \cup [3, 4)$



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